15 · Current status and future trends in kelp forest ecosystems

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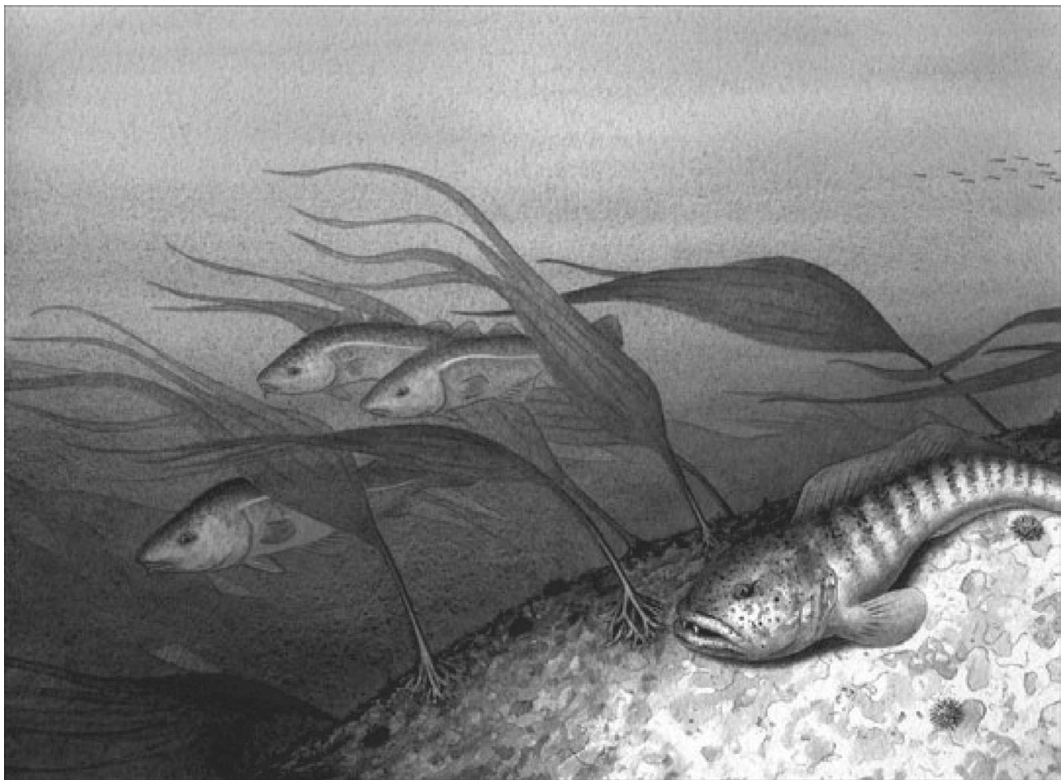

INTRODUCTION

Kelp forests dominate shallow rocky coasts of the world's cold-water marine habitats. They are comprised primarily of brown algae in the order Laminariales and produce one of the largest biogenic structures found in benthic marine systems. Kelp forest ecosystems include structure-producing kelps and their associated biota such as marine mammals, fishes, crabs, sea urchins, lobsters, molluscs, other algae and epibiota that collectively make this one of the most diverse and productive ecosystems of the world (Mann 1973, 2000; Dayton 1985*a*, *b*). Economically, kelp forest ecosystems have been significant to maritime peoples for thousands of years (Simenstad *et al.* 1978; Erlandson 2001). The aim of this chapter is to review how kelp forest ecosystems have changed at very large spatial scales over very long periods of time (decades to millennia). This perspective then provides a strong basis for assessing how different modern kelp ecosystems are from those in the

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past, and thus in turn allows an appreciation of possible future states.

LARGE-SCALE PATTERNS: GLOBAL INTERPLAY OF DIVERSITY, PRODUCTIVITY AND DISTURBANCE

Kelp forest ecosystems are distributed on rocky shores at mid-latitudes (Fig. 15.1). Taxonomically, the kelps are not diverse. The southern California coast of North America has the highest diversity of kelp; however, this comprises only 20 species of kelp distributed among 16 genera, most genera being monotypic. While total species diversity among kelp forest ecosystems can vary among taxonomic groups (for example, the algal diversity is greatest in south Australia: Bolton 1996), the species diversity of functionally important groups such as herbivores and large carnivores is greatest in southern California. In fact, North America provides excellent case studies for comparison because it has both the world's lowest- and highest-diversity kelp forests along its western North Atlantic and southern California coasts, respectively (Fig. 15.2).

The physical structure, algal biomass and organisms associated with kelp forests profoundly alter local and adjacent environments and ecologies. Kelp canopies dampen waves, and this influences water flow and associated processes of coastal erosion, sedimentation, benthic primary and secondary productivity, and recruitment of associated animals (Duggins *et al.* 1990; Mork 1996; Jackson 1998; Lvas & Trum 2001). The canopies reduce light, creating understorey conditions favourable for a suite of species adapted to low light intensity (Santelices & Ojeda 1984*a*) and thus potentially influencing interspecific competition among algae (Dayton 1985*a*). Kelps are substratum for numerous sessile animals and algae (Duggins 1980; Reed & Foster 1984; Dunton & Schell 1987) and habitat for mobile organisms specialized in living and feeding directly on the kelp or its associated assemblages. For example, trophically specialized limpets depend upon kelp for their existence (Steneck & Watling 1982; Estes & Steinberg 1988; Bustamante *et al.* 1995). Kelp forest architecture provides habitat, nursery ground and food for myriad mobile pelagic and benthic organisms (Bernstein & Jung 1980; Bologna & Steneck 1993; Levin 1994). Few fish species

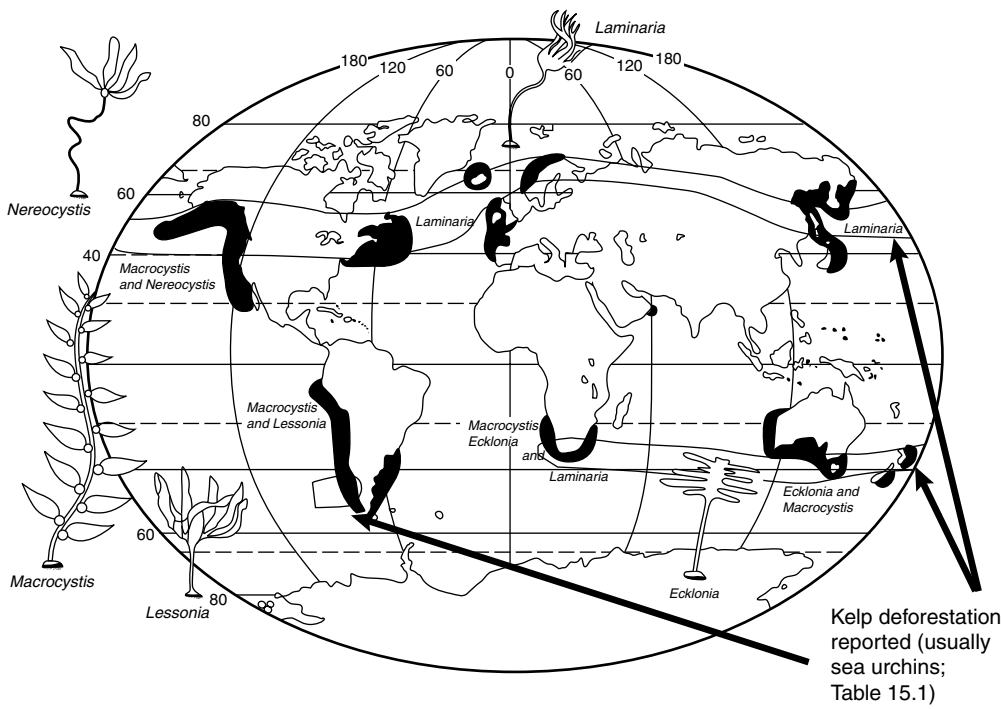

Fig. 15.1. Kelp forest distributions of the world and their dominant genera. Note mid-latitude bands in both hemispheres where kelp deforestation by sea urchins is common. (From Raffaelli & Hawkins 1996, modified by Steneck *et al.* 2002.)

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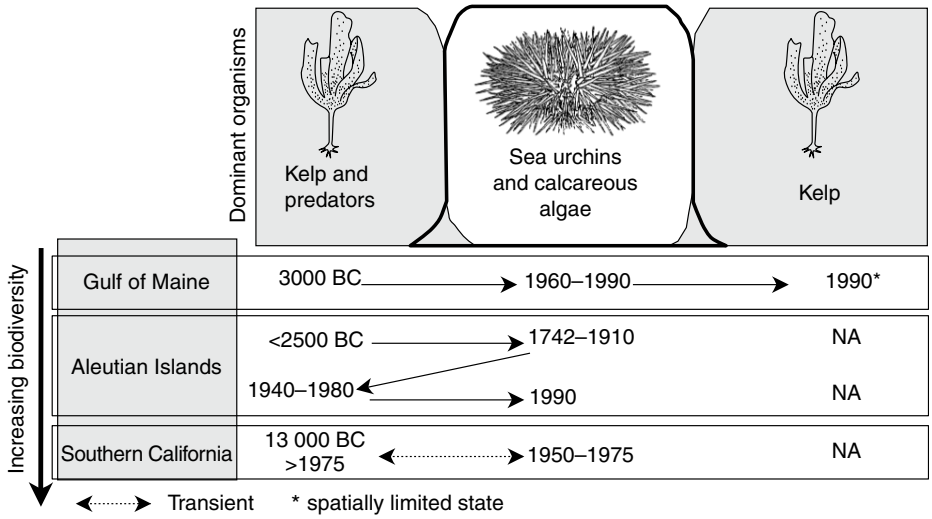

Fig. 15.2. Timing of phase changes in community state of kelp forests of North America. Kelp with vertebrate predators, sea urchins without kelp and kelp without predators have been identified for some or all of the case-study locations. Kelp forests are listed from the greatest number of trophic levels on the left to fewest trophic levels on the right. Case studies are listed from lowest species diversity in Maine to highest diversity in southern California. (From Steneck *et al.* 2002.)

feed directly in kelp (but see Andrew & Jones 1990; Jones & Andrew 1990); however, algal canopies serve as a nursery habitat for a range of rocky reef fishes feeding on microinvertebrates (Jones 1988; Carr 1989; Holbrook *et al.* 1990). Since predatory fishes use canopies as habitat, canopy loss can translate to increased survivorship of resident prey organisms and their larvae (Bray 1981; Gaines & Roughgarden 1987). Thus, if kelp canopies change, the ecosystem processes associated with them will be altered.

Kelps concentrate biomass and are a significant source of nutrition for coastal marine ecosystems via food webs based on macroalgal drift and detritus (Dunton & Schell 1987; Duggins *et al.* 1989). They commonly detach owing to the effects of storms and herbivores that damage kelp stipes, causing dislodgement. Usually, herbivores consume less than 10% of the living biomass (Mann 2000) thus leaving considerable detritus as food for benthic detritivores and microbes (Linley *et al.* 1981), which in turn make the carbon available to coastal suspension feeders. In effect, kelp forests concentrate and magnify secondary production, thereby supporting complex food webs in adjacent coastal zones (Duggins *et al.* 1989; Bustamante & Branch 1996; Mann 2000).

THE ECOLOGY OF KELP FORESTS

Kelp forests persist in a balance between ecological processes driving succession and stability (Dayton *et al.* 1984). They wax as a result of greater recruitment and net productivity, and wane with biomass loss from disturbances both physical and biological. Several strong ecological interactions control forest development and deforestation. The successional processes include spore dispersal, germination, gametogenesis, growth and intra- and interspecific competition. The deforestation also results from many ecological processes that can release populations of herbivores. All these processes operate at varying spatial and temporal scales and they will be the focus of this section, which will consider where kelp can live and develop forests, under what conditions kelp deforestation takes place, and what the out-of-system consequences of deforestation are.

Kelp forest distribution and development

Globally, kelp forests grow on shallow rocky shores in a mid-latitude band where light and oceanographic conditions allow the development and persistence of this growth form (Fig. 15.1). While kelps can grow in Arctic and sub-Antarctic regions (see Dunton & Dayton 1995), their

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abundance and diversity are low there (probably owing to light limitations: Dunton 1990; Henley & Dunton 1997) and thus they rarely develop forests above about 60° latitude. Similarly, warm temperatures and low nutrient concentrations generally prevent kelp forests from developing in subtropical or tropical regions (Bolton & Anderson 1987; Gerard 1997). The lowest-latitude kelp beds usually correspond with ocean-current anomalies in latitudinal gradients of warm temperatures and/or low nutrient conditions. For example, kelp forests are found near the tropics of Cancer and Capricorn only along the western coasts of southern California to Mexico, northern Chile to Peru and western South Africa (Fig. 15.1). In each of these cases, ocean currents, directed by Coriolis forces, flow toward the equator advecting cool and often nutrient-rich water. Low-latitude kelps (usually <40° latitude) are often diminutive and share or lose community dominance to fucoids such as *Sargassum* and other large brown algae that become more diverse and abundant toward the tropics (Hatcher *et al.* 1987).

In general, the three interacting processes that control the development of kelp forests are recruitment, growth and competition (North 1994). Locally, kelp forests are established and maintained by successful settlement of zygotes, which grow and are thinned by mortality from intraspecific competition during their benthic life (Dayton *et al.* 1984; Reed & Foster 1984; Chapman 1986). Recruitment is often seasonal and influenced by environmental conditions at the time of settlement. In complex kelp forests tiered with multiple levels (for example canopy, stipitate and prostrate forms: Steneck *et al.* 2002), such as the California kelp forest, kelp recruitment and growth is regulated by light available through breaks in the kelp canopy (Dayton *et al.* 1984; Reed & Foster 1984; Graham *et al.* 1997) as well as by available nutrients (Dayton *et al.* 1999). Following intense storms that deforest or thin kelp canopies, recruitment is usually strong, but the kelp species that grow to dominance will depend upon nutrient conditions at the time (North 1994; Tegner *et al.* 1997).

Kelp growth depends on interactions among nutrient availability, temperature and light. Kelps dominate cold-water coastal zones (Fig. 15.1) but can become physiologically stressed at high sea temperatures, particularly when nutrient availability is low (Tegner *et al.* 1996*a*; Gerard 1997). In some regions without upwelling, periods of low nutrient concentrations correspond with warm summer temperatures when the water is stratified (Chapter 14). The combined effects of low nutrients and high rates of respiration result in kelp plants that erode more rapidly than they grow (Gagne *et al.* 1982; R. S. Steneck, unpublished data 1984). In kelp forests driven by the upwelling of new nitrogen, such as those of southern California, warm surface water is a surrogate for low nutrient availability (Tegner *et al.* 1996*a*). In this system, when El Niño events disrupt coastal upwelling, kelp plants become nutrient-starved and die back (Tegner & Dayton 1991). As a result, the distribution, abundance and size of kelp plants decline as sea-surface temperatures increase (Dayton *et al.* 1999).

Kelp deforestation

Widespread kelp deforestation can result from disease, herbivory and physiological stress, or interactions among those processes. In lower-latitude (usually <40°) kelp forests, periodic deforestation results from oceanographic anomalies in temperature, salinity or nutrients that either kill kelp directly or trigger diseases that become lethal to physiologically stressed plants. At mid-latitudes (*c.*40–60°), herbivory from sea urchins is the most common and important agent of kelp deforestation in both hemispheres. Latitudinal differences in patterns and processes shaping kelp forests have resulted in different researchers working in the same kelp forest system but reaching different conclusions (Foster 1990). The geography of kelp deforestation patterns and processes will be addressed here.

Kelp-free patches have probably always occurred at some scale, but those created by physical factors tend to be relatively small and short-lived. The oldest term for algal deforestation is the Japanese word *isoyake*, which literally means 'rock burning' (D. Fujita, personal communication 2002). The word was coined by Yendo (1902, 1903) to describe algal deforestation in coastal zones of central Japan where the algal decline was thought to have resulted from salinity anomalies (Yendo 1903, 1914) rather than grazing because herbivorous sea urchins were rare. The *isoyake* killed all foliacious algae first and then all encrusting coralline algae before recovering several years later (Yendo 1914). Other mass mortalities of *Ecklonia*- and *Eisenia*-dominated kelp forests resulted from incursions of the Kuroshio Current along the central Japan coast (D. Fujita, personal communication 2002). On Honshu Island, near the southern limit of Japanese kelp, anomalous incursions of the warm Tsushima Current periodically create *isoyake* conditions. Such oceanographically induced kelp deforestations are usually short-lived and reversible, as was the first described *isoyake* case in Japan (Yendo 1914).

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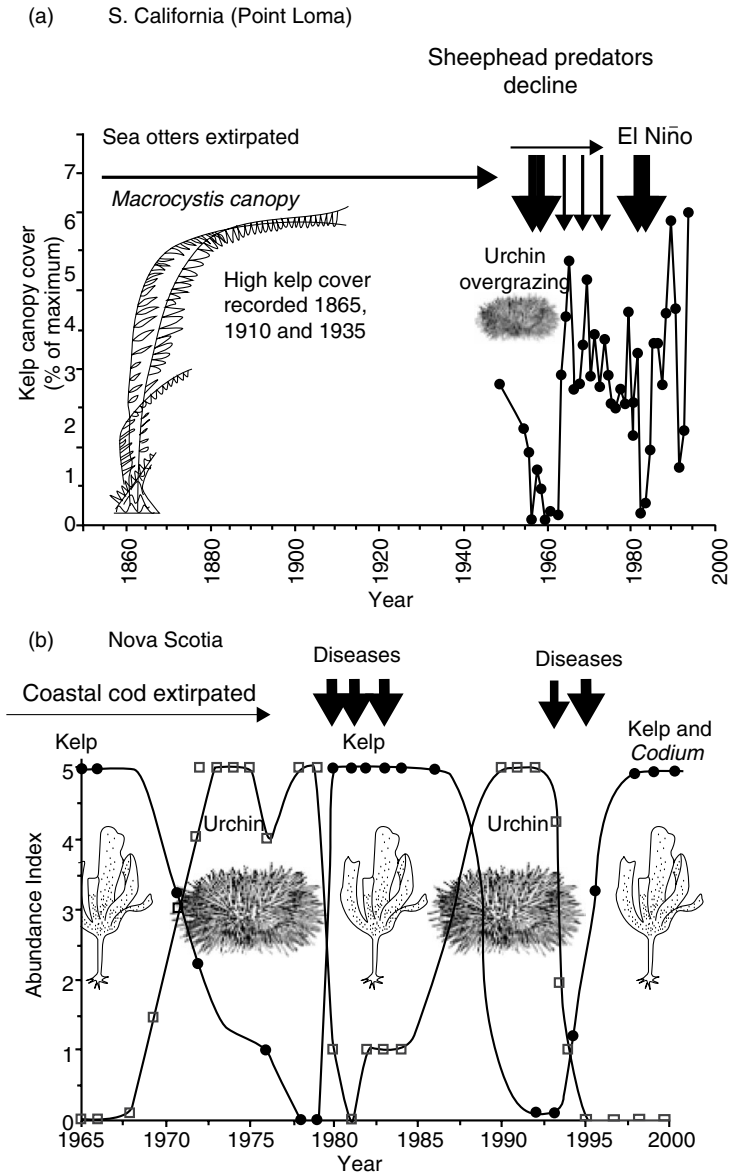

Fig. 15.3. Temporal trends in kelp forested and urchin-grazed deforested periods in (a) Southern California and (b) Nova Scotia. Arrows indicate the timing and magnitude of change in major community-altering forcing functions. Width of arrowheads indicates magnitude of the forcing function's impact. The strong fluctuations in Nova Scotia's sea urchin population result from periodic disease-induced mass mortalities. Kelp abundance fluctuates inversely with urchin abundance (Scheibling *et al.* 1999). (From Steneck *et al.* 2002.)

Kelp deforestations also result from El Niño events. Strong El Niños halt coastal upwelling of nutrient-rich water and cause surface waters to warm (Dayton *et al.* 1992; Chapter 14). These anomalies in California (Fig. 15.3a) have caused patchy deforestation followed by rapid recovery (Tegner & Dayton 1987; Dayton *et al.* 1992). Such physiological stresses are likely to be more common towards the low-latitude limits of kelp ranges. For example, the

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northern limit of three species of brown algae in northern Chile shifted south towards higher latitudes following the El Niño event of 1982/3 (Peters & Breeman 1993). Such stresses may make kelps more susceptible to disease. Low-latitude kelp in northern New Zealand succumbed to a disease, which may have resulted from a physiological stress (Cole & Babcock 1996; Cole & Syms 1999).

Within mid-latitudes, where kelp development is less likely to be limited by physical processes such as temperature, nutrients and light, deforestation most often results from sea-urchin herbivory (Leighton *et al*. 1966; Lawrence 1975; Himmelman 1980; Dayton 1985*a*, *b*; Duggins 1980; Estes & Duggins 1995; Mann 2000; Steneck *et al*. 2002). This is most evident in the northern hemisphere where the most widespread and long-lasting herbivore-induced kelp deforestations have resulted from sea-urchin grazing (Table 15.1). These primarily *Laminaria*-dominated kelp forests (Fig. 15.1) have been reduced in historical times to coralline-dominated 'urchin barrens' (urchin-induced kelp deforestation) in the Aleutian Islands of Alaska, the Gulf of Maine, Canadian Maritimes (to Newfoundland: Himmelman 1980), northern Japan (Hokkaido Island), Iceland and northern Norway (Table 15.1). South of those regions, forests either remain intact such as in southern California (Fig. 15.3a; Dayton *et al*. 1984) or are deforested patchily in relatively few regions such as southern Norway (Sivertsen 1997), Ireland (Kitching & Thain 1983), Britain (Kain 1975) and southern Japan (Honshu Island: Fujita 1998).

Kelp deforestation from sea urchins is less common in the southern hemisphere (Table 15.1). In Chile, there has been extensive sea-urchin deforestation in the latitude range 46°–54° S (Dayton 1985*b*), while only patchy deforestation has been reported in north central Chile (i.e. 32° S: Ojeda & Santelices 1984) (Table 15.1). Dense *Macrocystis* forests with few urchins have been described in the southernmost regions of South America (*c*.55° S), in Chile (Castilla & Moreno 1982; Santelices & Ojeda 1984*b*), and in Argentina (Dayton 1985*b*). While southernmost Chile has four sea urchin species (*Loxechinus albus, Pseudechinus magellanicus, Arbacia dufresnei* and *Austrocidaris canaliculata*), their densities are low and they subsist on drift kelp and rarely graze attached *Macrocystis* plants (Dayton 1985*b*). Along the east coast of southernmost South America to the northern limit of kelp in Argentina (42° S: Barrales & Lobban 1975), the sea urchin *Arbacia dufresnei* is the dominant herbivore but its grazing impact on kelp forests is minimal (Barrales & Lobban 1975).

Elsewhere in the southern hemisphere, sea-urchin deforestation is patchy or restricted to particular depth zones. In New Zealand, a band of sea urchin 'barrens' exists at intermediate depths at around 10 m (Choat & Schiel 1982). Urchins there may be prevented by wave turbulence from grazing shallower zones. In South Australia, kelp deforestation is restricted to regions of high spatial heterogeneity that provide shelter for nocturnally grazing sea urchins (Andrew 1993). Recent *Macrocystis pyrifera* deforestation in Tasmania, largely affected by the past El Niño events, has resulted in an area reduction of at least 50% over the last 10 years (Edyvane 2003). At the same time, high-density populations of the 'range-expanding' sea urchin *Centrostephanus rodgersii* have become established, increasing grazing pressure and forming extensive 'barrens', while the alien Japanese kelp *Undaria pinnatifida* now occupies regions of the former kelp forest habitat (Edyvane 2003; Valentine & Johnson 2003). In South Africa, sea urchins alone do not overgraze kelp forests (Velimirov *et al*. 1977), but grazer-induced phase shifts can occur from a diversity of herbivores (G. M. Branch, personal communication 2001).

Whatever regulates sea-urchin abundances or grazing behaviour also often controls the distribution and abundance of kelp forests (Steneck *et al*. 2002) (Table 15.1). Predators are commonly strong interactors (*sensu* Paine 1980) and, as such, are the single most important agent controlling sea-urchin populations (Dayton 1975; Cowen 1983; Duggins 1983; Tegner & Levin 1983; Estes & Duggins 1995; Sala *et al*. 1998; Steneck 1998; Shears & Babcock 2002). When sea-urchin predators are dramatically reduced or eliminated through intense and unsustainable fishing, subsequent hyperabundance of the sea urchins is common, and kelp deforestation results (Lawrence 1975; Estes & Duggins 1995; Steneck 1998). Exceptions to this pattern are found at lower latitudes where diverse guilds of kelp forest herbivores and predators compensate for the loss of single predator species. Sea-urchin abundance can also be influenced by other factors such as disease (Fujita 1998; Scheibling *et al*. 1999), turbulence (Choat & Schiel 1982) and storms (Dayton 1985*a*; Ebling *et al*. 1985) that can locally or periodically reduce sea-urchin abundance and thus control kelp forest development indirectly (Foster & Schiel 1985).

Widespread, long-lasting kelp deforestation from sea urchins may be a relatively recent phenomenon. In the Aleutians of Alaska, the transition may have occurred

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Table 15.1. *Comparison of subtidal kelp forest ecosystems of the world*

	Kelps and their controlling agents[a]				Spatial and temporal scale of deforestation				
Site and latitude	Dominant kelps	Deforesting herbivores	Predators or diseases of kelp herbivores	Drift kelp, disease, oceanography	Regional distribution	Depth range	Local distribution	Duration deforested	References
Western North Atlantic									
Nova Scotia 43–45° N	*Laminaria* (1), *Agarum* (1)	Echinoid (1)	Fishes (2), urchin disease	Drift, disease	Widespread	Broad	Homogeneous	Decades	Steneck *et al.* (2002)
Maine 43–44° N	*Laminaria* (1), *Agarum* (1)	Echinoid (1)	Fishes (2), (crabs)		Widespread	Broad	Homogeneous	<Decades	Steneck *et al.* (2002)
Eastern North Atlantic									
North Iceland 65° N	*Laminaria* (1)	Echinoid (1)	?		Widespread	Broad	Homogeneous	?	Hjorleifsson *et al.* (1995)
North Norway 65–71° N	*Laminaria* (1)	Echinoid (2)	Seabirds		Widespread	Broad	Homogeneous	Decades	Hagen (1983), Sivertsen (1997), Bustnes *et al.* (1995)
South Norway 55–64° N	*Laminaria* (1)	Echinoid (1)	?		Restricted	Broad	Patchy	Decades	Sivertsen (1997)
Britain and Ireland 52–55° N	*Laminaria* (3)	Echinoids (2)	Crabs		Restricted	Broad	Patchy		Kain (1975), Kitching and Ebling (1961), Ebling *et al.* (1966)
Eastern North Pacific									
Alaska (Aleutians) 50–55° N	*Alaria* (1), *Laminaria* (3), *Thalassiophyllum* (1), *Agarum* (1)	Echinoid (1)	Sea otter		Widespread	Broad	Homogeneous	>Decades	Steneck *et al.* (2002)
Southern California 30–35° N	*Macrocystis* (1), *Laminaria* (1), *Pterygophora* (1)	Echinoids (3), gastropods (8), fishes (2)	Sea otter (1), fish (1), lobster (1)	Drift, oceanographic (ENSO) events	Restricted	Broad	Patchy	<Decade	Steneck *et al.* (2002)

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Western North Pacific									
North Japan (SW Hokkaido) 39–46° N	*Laminaria* (2)	Echinoids (1–3)	Crabs, urchin disease	Oceanogaphic event	Widespread	Broad	Homogeneous	Decades	Fujita (1998), D. Fujita, personal communication (2001)
South Japan (W. Honshu) 36–38° N	*Undaria* (1), *Eisenia* (1), *Ecklonia* (1)	Echinoids (3) , fish (1)	?	Oceanographic event	Restricted	Broad	Patchy	?	Fujita (1998), D. Fujita, personal communication (2001)
Eastern South Pacific									
North Chile 18–42° S	*Lessonia* (1), *Macrocystis* (1)	Echinoids (2), fishes (1), gastropods (2)	Asteroids (3), Fishes (3)	Drift	Widespread	Shallow	Patchy	Decades	Vasquez (1993), Vasquez and Buschmann (1997), Ojeda and Santilices (1984)
South Chile 46°–54° S	*Macrocystis* (1), *Lessonia* (2)	Echinoids (1), gastropods (1)	Asteroids (1)	Drift	Restricted	Shallow	Patchy	Decades	Dayton (1985b)
Southern-most Chile 55° S	*Macrocystis* (1), *Lessonia* (2)	Echinoids (4)	Asteroids (1)	Drift	None				Castilla and Moreno (1982), Santelices and Ojeda (1984b), Vasquez *et al*. (1984)
Argentina 42–55° S	*Macrocystis* (1), *Lessonia* (1)	Echinoid (1)	?	?	None				Barrales and Lobban (1975)
Western South Pacific									
Australia (New South Wales) ~32–35° S	*Ecklonia* (1)	Echinoids (1), fishes (1)	Fishes (2)		Widespread	Moderately deep	Patchy	?	Andrew (1993), Andrew and Underwood (1993), Andrew (1994), Andrew and O'Neill (2000)
Australia (Tasmania) ~43° S	*Macrocystis* (1), *Ecklonia* (1)	Echinoid (1)	Fish (1), lobster (1)		Restricted	Broad	Homogeneous	Years	Edyvane (2003)

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Table 15.1. (*cont.*)

	Kelps and their controlling agents[a]				Spatial and temporal scale of deforestation				
Site and latitude	Dominant kelps	Deforesting herbivores	Predators or diseases of kelp herbivores	Drift kelp, disease, oceanography	Regional distribution	Depth range	Local distribution	Duration deforested	References
New Zealand (North Island) 34–37° S	*Ecklonia* (1), *Lessonia* (1)	Echinoids (2), gastropods (2)	Fishes (?), lobster (1)	Kelp disease	Widespread	Mid-depth	Homogeneous	Decades	Choat and Schiel (1982), Andrew and Choat (1982), Choat and Ayling (1987), Schiel (1990), Cole and Babcock (1996), Babcock *et al.* (1999), Cole and Syms (1999)
New Zealand (South Island) 41–47° S	*Ecklonia* (1), *Lessonia* (1), *Macrocystis* (1)	Echinoids (1)	?		Restricted	Broad	Patchy	?	Schiel (1990), Schiel *et al.* (1995)
Eastern South Atlantic									
South Africa ~30–35° S	*Ecklonia* (1), *Laminaria* (1), *Macrocystis* (1)	Echinoids (1), gastropods (1)	Lobster (1), fish (?)		Widespread	Deep only	Patchy	?	Anderson *et al.* (1997), G. Branch, personal communication (2001)
Eastern Indian Ocean									
W. Australia 28° S	*Ecklonia* (1)			Oceanography (high temperature and low nutrients)					Hatcher *et al.* (1987)

[a] Numbers in parentheses denote number of ecologically important species in subtidal kelp forests for specified taxa.
Source: After Steneck *et al.* (2002).

early in the twentieth century (Estes & Duggins 1995). In Japan, fishers observed deforestation and patches of corallines first in the early 1930s (Fujita 1987, 1998), although sea urchins were not mentioned at that time. Later, enlarging coralline patches grazed by sea urchins were reported during the 1950s and 1960s (Ohmi 1951; Fujita 1998). In California during the 1960s, the term 'barrens' was coined to describe urchin-induced kelp deforestation (Leighton *et al.* 1966). In the North Atlantic, the first gaps in kelp forests were reported in the 1960s for the Gulf of Maine (Lamb & Zimmerman 1964), Nova Scotia (Edelstein *et al.* 1969; Breen & Mann 1976), Ireland (Ebling *et al.* 1966) and the British Isles (Jones & Kain 1967). Gaps in kelp forests in the western North Atlantic coalesced and expanded during the 1970s and 1980s in Nova Scotia (Mann 1977) and the Gulf of Maine (Steneck 1997). Expansive coralline 'barrens' existed in Newfoundland in the late 1960s (Himmelman 1980) and were possibly present there earlier (Hooper 1980). In the eastern North Atlantic, widespread urchin-induced deforestation was first observed in northern Norway in the early 1980s (Hagen 1983; Sivertsen 1997) and in the early 1990s in Iceland (Hjorleifsson *et al.* 1995). By the mid 1970s, sea urchins were viewed as the major cause of kelp deforestation (Lawrence 1975) such that by the mid 1980s conferences were held to discuss (among other things) how sea urchins could be eradicated (Pringle *et al.* 1980; Bernstein & Welsford 1982). Urchin-induced kelp deforestation was widely reported in mid-latitudes of the northern hemisphere from 40° to 60° N (higher in the eastern North Atlantic owing to the Gulf Stream) during the 1960s and 1980s. At the time, some researchers openly wondered if kelp deforestation was an 'irreversible degradation' (Mann 1977).

Biodiversity, trophic cascades and rates and consequences of kelp deforestation

In the North American case studies, the extirpation of predators led to increased herbivory by sea urchins resulting in kelp deforestation at local to widespread spatial scales (Fig. 15.3b). In the western North Atlantic and Alaska, where predator diversity is low (Steneck *et al.* 2002), the transition between kelp forests and coralline communities was rapid, frequent (Fig. 15.3b), widespread and in some cases long-lasting (Tables 15.1 and 15.2). These patterns differ from southern California, where the diversity of predators, herbivores and kelps are high (Steneck *et al.* 2002), deforestation events have been rare or patchy in space and short in duration (Harrold & Reed 1985), and no single dominant sea-urchin predator exists (Tegner & Dayton 1997) (Fig. 15.3a). The biodiversity within functional nodes, such as trophic levels, is critical to the structure and functioning of kelp forest ecosystems. Nevertheless, even the most diverse systems can lose, and are losing, their functional diversity as overfishing reduces the ecologically effective population densities of important species, rendering them ecologically extinct (Estes *et al.* 1989). All of this suggests that the fragility and rate of change in kelp forest ecosystems may depend on local biodiversity. It remains an open question whether diverse kelp forests will persist or if serial disassembly and instability will inevitably result.

The direct consequences of kelp deforestation among kelp-dependent communities will be profound. In southern Africa, where extensive forests of the kelps *Ecklonia maxima* and *Laminaria pallida* exist (Velimirov *et al.* 1977; Field *et al.* 1977, 1980) (Fig. 15.1), kelp is the principal primary producer contributing to the energetics of the shallow rocky subtidal food webs (see Field *et al.* 1977; Newell *et al.* 1980; Wickens & Field 1986). In addition, kelp decay plays a significant role for microheterotrophs through the regeneration of nutrients available to a range of subtidal producers and consumer species (for example Steele 1974; Fenchel & Blackburn 1979; Newell *et al.* 1988; Painting *et al.* 1992). Kelp deforestation can also affect surrounding marine and terrestrial communities (Graham *et al.* 2003, box 2). Drift from giant kelp (*Macrocystis pyrifera*) contributes 60–99% of beach-cast autotrophic detritus in the southern California bight (Zobell 1971) and similar amounts in eastern Nova Scotia (Mann 2000). Offshore contributions are facilitated by gas-filled floats and stipes which, when adult sporophytes are detached from the bottom due to grazing or physical disturbance, provide for long-distance dispersal by rafting (Harrold & Lisin 1989; Hobday 2000). When floating kelp rafts are deposited on the shore, the floats break, and they wash into shallow nearshore habitats and ultimately into offshore basins (Graham *et al.* 2003). Secondary productivity of both shallow (Vetter 1995) and deep-sea (Harrold *et al.* 1998) soft-sediment systems is consequently driven in a large part by allochthonous food subsidies from regional kelp resources (e.g. Chapters 17, 19 and 22). Kelp detritus can also make its way into nearby intertidal food webs through the capture either of fine kelp particles by filter-feeders such as mussels (Duggins *et al.* 1989) and clams

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Table 15.2. *Spatial and temporal scale of change in kelp forest ecosystems in three regions of North America*

Event	North-west Atlantic	West Aleutian Islands	Southern California
Pristine state (prior to human contact)	Kelp forest	Kelp forest	Kelp forest
Scale (patch size)	100 > 500 km (Steneck *et al.* 2002)	200–1000 km (Estes *et al.* 1989)	<10 km (Harrold and Reed 1985 and Tegner *et al.* 1996a)
Human contact/occupation	10 000 ybp (Bourque 1995)	8000 ybp (Simenstad *et al.* 1978)	12 000–13 000 ybp (Erlandson *et al.* 1996)
Marine organisms in diet	5000 ybp (Bourque 1995)	4500 ybp (Simenstad *et al.* 1978)	11 600 ybp (Erlandson *et al.* 1996)
First known phase change	?–40 ybp (Adey 1964)	2500 ybp (Simenstad *et al.* 1978)	4000–6000 ybp (Erlandson *et al.* 1996, Erlandson *et al.* 2004)
First European contact/ exploitation	460/400 ybp (Steneck (1997)	260/260 ybp (Simenstad *et al.* 1978)	460/200 ybp (Simenstad *et al.* 1978)
Kelp bed re-establishement rate	0.5–4 yrs (Steneck *et al.* 2002, Jones and Kain 1967)	2 yrs (Estes *et al.* 1989)	0.5 yrs (Harrold and Reed 1985 and Tegner *et al.* 1996a)
Alternate (kelp-free) state	Coralline/urchin	Coralline/urchin	Coralline/urchin
Storms remove dominant kelp	Small scale (Steneck *et al.* 2002)		Large scale (Dayton *et al.* 1992)
Recent apex predators	Crabs (Leland 2002)	Killer whales (Estes *et al.* 1998)	
Introduced competitors	Bryozoan and *Codium* (Lambert *et al.* 1992)		

ybp, years before present
Source: After Steneck *et al.* (2002).

(Soares *et al.* 1997) or of large pieces of drift kelp by limpets (Bustamante *et al.* 1995) and sea urchins (Day & Branch 2002; Rodriguez 2003; Chapter 14). All these kelp consumers are in turn strong interactors in their respective food webs and communities, and any perturbation to their populations will alter their community structures. Kelp detritus enhances the inherently low productivity of terrestrial ecosystems on arid islands (Polis & Hurd 1996). During dry years, carbon and nitrogen from marine bird and mammal faeces and beach-cast marine detritus fuel terrestrial productivity, with the greatest impact on islands with large shoreline-to-area ratios (Graham *et al.* 2003). The importance of marine subsidies declines during rainy years when high precipitation increases terrestrial production. Excellent examples are found in southern California where numerous low-productivity islands (such as the Channel Islands) are imbedded within a highly productive marine system. In addition to localized areas of high guano and pinniped excrement accumulation, the shoreline is loaded with large quantities of kelp detritus (Graham *et al.* 2003). All of this suggests that changes in the volume of organic material exported from kelp forests in temperate arid islands and coasts (for example western coasts of America and Africa) will probably affect the community structure and composition of intertidal rocky shores (Chapter 14), sandy beach infauna (Chapter 17), coastal food webs (e.g., Chapters 11–13) and benthic assemblages in deep-water canyons (Chapter 22).

Few have examined the consequences of declining kelp for the biodiversity of mobile organisms using these habitats. Responses of kelp-associated fishes indicate the potential ecological price that the 'habitat-user' pays for habitat degradation. While some fishes play key functional roles as ecosystem 'drivers', by far the greatest biodiversity

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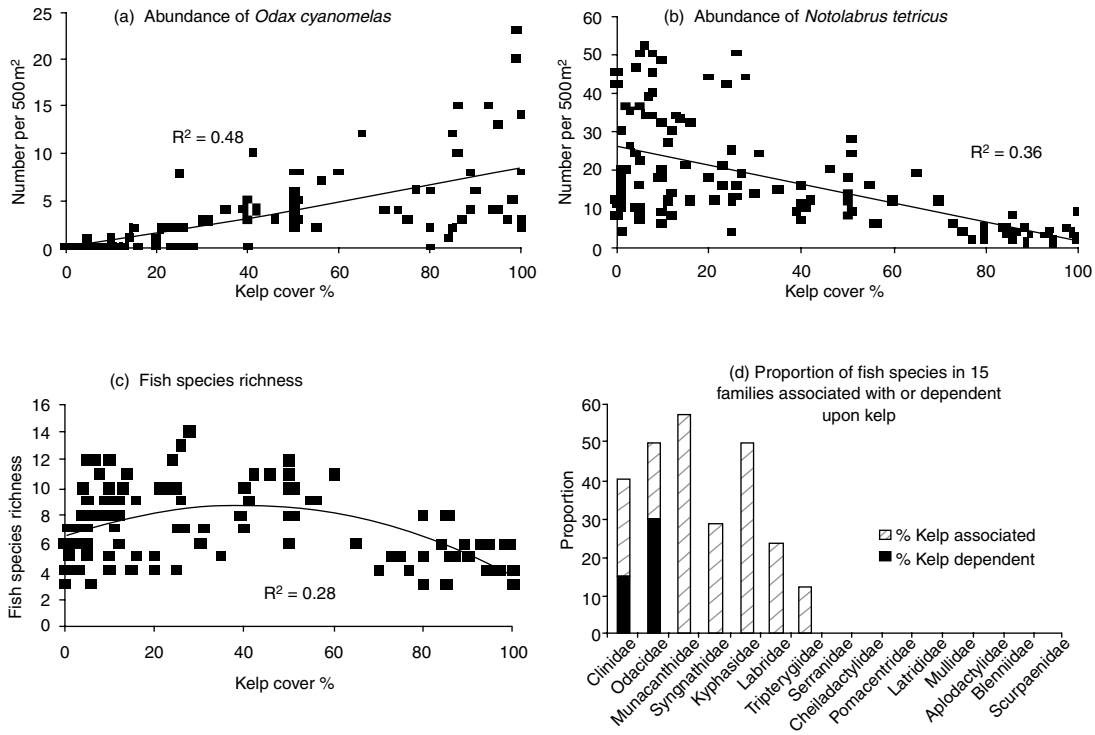

Fig. 15.4. Fish associations with kelp forests in Victoria (Australia): (a) kelp-feeding fish *Odax cyanomelas*, (b) turf-feeding wrasse *Notolabrus tetricus*, (c) fish species richness as a function of kelp cover and (d) proportion of fish species associated with or dependent upon kelp. (From G. P. Jones, unpublished data 2001.)

of fish species are ecosystem 'passengers'. Rocky-reef fish assemblages typically comprise species that are a mix of those associated with kelp stands and those associated with open reef areas, which are either urchin-grazed or dominated by algal turfs. The biogenic structure of habitats has a substantial influence on associated fish assemblages (Jones 1988; Carr 1989; Holbrook *et al.* 1990, 1993; Jones & Andrew 1993; Syms & Jones 1999), and the abundances of individual passenger species are often explained by the availability of their preferred habitats (see Kingett & Choat 1981; Bodkin 1988; Anderson 1994; Levin & Hay 1996; Hartney & Grorud 2002). For example, on the southern coast of Victoria (Australia) the abundance of the kelp-feeding fish *Odax cyanomelas* is positively correlated with kelp cover, while that of the wrasse *Notolabrus tetricus*, which is associated with algae turfs, is negatively correlated with kelp (Fig. 15.4 a, b). Processes that maintain spatial and temporal heterogeneity in kelp-dominated habitats will be vital to maintain fish biodiversity. Fish species richness is often correlated with habitat complexity (Angel & Ojeda 2001; García-Charton & Pérez-Ruzafa 2001). In the case of southern Australia, greatest fish diversity occurs at sites of intermediate kelp cover where patch complexity is the greatest (Fig. 15.4c). Any process resulting in a homogeneous habitat structure, whether because of total deforestation or invasive species (Levin *et al.* 2002b), will be likely to have a negative impact on reef fish biodiversity (Harmelin-Vivien *et al.* 1999). Of 237 fish species across 15 families in southern Australia, only seven species (3%) may be totally dependent on kelp for food or living space, including one herbivore and a few species closely associated with kelp canopies (Fig. 15.4d). However, a further 41 species (17%) are strongly kelp-associated, using kelp as their primary recruitment habitat. Thus, while total deforestation will probably result in population declines and some local extinctions, the major effect of a phase shift from kelp-dominated to deforested habitat will be a phase shift in the structure of fish communities.

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POTENTIAL STATES IN 2025

Extrapolation of known trends

It is likely that climate change, human population growth, coastal development, oil spills, fisheries-induced impacts, disease and invasions of non-native species will continue and possibly accelerate over the 2025 time horizon. All of these may well contribute to a continuing disassembly of kelp forest ecosystems. It is difficult to extrapolate known trends into the future, because non-linear thresholds and complex interactions can cause ecosystems to behave unpredictably (e.g. Scheffer *et al.* 2001). Some activities may change the nature of functional relationships. For example, as discussed, overfishing of sea-urchin predators can cause kelp forests to decline, but overfishing of sea urchins themselves can have the reverse effect.

It is tempting to speculate that if sea-urchin populations usually increase as their predators are extirpated, then perhaps the sea-urchin fishery exploitation will expand globally and reverse these trends. However, fishing pressure on sea urchins hinges on the marketability of the urchin roe. Since the primary market for sea-urchin roe is Japan along with a growing list of primarily Asian countries, often the only way harvesters can make a profit collecting these animals, which usually contain no more than 15% usable product by volume, is to receive a high price per unit mass. In the USA, the domestic price per unit mass would be too great to make the fishery viable. Thus foreign currency exchange rates functionally regulate fishing pressure on this driver of many kelp forest ecosystems. However, because it is impossible to predict global currency exchange rates, it will be equally difficult to predict the fishing pressure on sea urchins over the next several decades.

While global temperature is expected to increase to the year 2025, related patterns of droughts, fires, heatwaves, storms and precipitation are expected to increase in some areas but decline in others (Houghton *et al.* 1996). This underscores the limitations of attempting to generalize about kelp forest ecosystems globally. However, the three cases described offer insights into risks that may in future befall these ecosystems with widely varying diversities and scales in space and time. This review will now consider trends from the largest spatial and temporal scales to the smallest.

Ocean-climate change: global warming, regime shifts and ENSO

Ocean temperature regulates the physiology and biogeography of marine algae (Adey & Steneck 2001). Global warming, regime shifts, and El Niño–Southern Oscillation (ENSO) events are climate-driven thermal effects that can variably impact kelp forest ecosystems at a wide range of temporal and spatial scales. Global warming operates at the largest temporal and spatial scales, but the projected changes to the 2025 time horizon are modest compared to regime shifts at the scale of whole ocean basins. Regime shifts cause temperature fluctuations nearly an order of magnitude greater and persist at the temporal scale of decades (Steneck *et al.* 2002). Superimposed on both of those changes are ENSO events, which can cause the greatest temperature anomalies but impact coastal zones at smaller scales and over periods of only a year or two. The interaction among these three ocean climate effects is complicated because each varies differently in space and time. However, when two or more thermal anomalies coincide, the compounded perturbations to kelp forests can be very substantial (Paine *et al.* 1998).

The rate and magnitude of global temperature rise during the past century is unprecedented over at least the past 600 years (Houghton *et al.* 1996; Mann *et al.* 1998). As a group, kelps are limited to cold-water coastal zones (Fig. 15.1). Thus, kelp living close to their upper thermal limits will be likely to recede to higher latitudes during protracted warming periods.

Temperature effects are complicated by a variety of ocean–atmosphere interactions. While the long-term frequency of strong storms in the eastern North Pacific Ocean has changed relatively little since the year 1625 (Enfield 1988), recently the frequency and intensity of extreme cyclones has increased markedly (Graham & Diaz 2001). Two storms in the 1980s were described as 'storms of the century' (Seymour *et al.* 1989). The recent steady (rather than stepwise) increase in storm activity may be related to rising sea-surface temperatures in the western tropical Pacific (Graham & Diaz 2001). Thus, abiotic storm-induced disturbances of kelp forests could increase commensurately with global warming and would be likely to affect negatively the upper and nearshore local limits of kelps (Graham 1997). However, kelp forests usually recover rapidly from such disturbances (Tegner *et al.* 1997).

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In 1982/3, a 4–5 °C ENSO warming halted coastal upwelling and created one of the most severe disturbances of a giant kelp forest ever documented (Paine *et al.* 1998) (Fig. 15.3a). Nitrogen concentrations in seawater influence the welfare of kelp and vary with ocean temperature (Gerard 1997). When surface waters become stratified owing to ENSO or as might be expected with global warming, nitrogen concentrations decline and kelps become nitrogen limited and may die (Dayton *et al.* 1999). *Macrocystis* beds are particularly vulnerable because they possess limited nitrogen storage capacity (Gerard 1982). Although groundwater and atmospheric inputs of nitrogen to coastal oceans have increased, the impact is modest or undetectable relative to natural fluxes in all but enclosed basins with limited flushing (Jickells 1998). Thus, trends towards increasing temperature and decreasing nitrogen availability in kelp forests are likely to continue.

The combination of warming coastal oceans and increased stratification will be likely to shrink the geographical range of kelp beds living closest to the tropics. Indirect effects of temperature on species that influence kelp forests will be discussed below. Sea-level rise of 20 cm by the year 2050 as a result of global warming (Houghton *et al.* 1996) could cause increased coastal erosion and turbidity which then could affect kelp depth distribution (Vadas & Steneck 1988) and possibly other processes such as photosynthesis and recruitment (see Graham 1996). Increased sedimentation from erosion could also reduce the area of substrate available for kelp settlement, as is currently postulated in Tasmanian kelp declines (Edyvane 2003).

Regime shifts and ENSO events can cause thermal anomalies of shorter duration but potentially greater impact. In general, ENSO events create warmer sea temperatures along the southern California coast, but the events are somewhat more frequent during regime shifts when warmer conditions prevail on that coast (Steneck *et al.* 2002). El Niño warming is usually followed by La Niña cooling (McGowan *et al.* 1998) and associated violent storms (Tegner & Dayton 1987, 1991; Seymoure *et al.* 1989; Tegner *et al.* 1997).

One of the best-studied climatic regime shifts occurred in the North Pacific in the mid 1970s, and the regime apparently ceased at the end of the 1990s (Steele 1998). If the duration of the post-1999 regime is similar to the last two, then cooler than average temperatures and perhaps lower than average El Niño frequency (and higher frequencies of La Niña and storm events) may be expected over much of the next quarter century (McGowan *et al.* 1998; Fei Chai, personal communication 2001). However, the duration of storm impacts in California is relatively brief and the ecosystem recovery rapid, so storms should have little lasting impact (Tegner *et al.* 1997) (Fig. 15.3a).

Changing coastal biodiversity: new apex predators and competitors

Changes in kelp forest biodiversity that affect functional components of kelp beds can disrupt the system in both predictable and unpredictable ways. The most conspicuous changes to the kelp forest result from overfishing key drivers such as apex predators and sea urchins (Estes & Duggins 1995; Jackson *et al.* 2001). Significant reductions in either can lead to trophic-level dysfunction, and alternate stable states or large-scale instabilities (Steneck *et al.* 2004). For example, extirpation of sea-urchin predators led to hyperabundances of sea urchins that had been stable for decades in Alaska (Steneck *et al.* 2002) and remarkably unstable in Nova Scotia (Fig. 15.3b) owing to epizootic disease cycles (Scheibling *et al.* 1999). Arguably, predator loss led to hyperabundances of species, setting the stage for disease-related mass mortalities (McNeill 1976).

Apex predators in pristine kelp forests were probably relatively large vertebrates in the northern hemisphere, while large invertebrates dominated many southern hemisphere kelp forests. The case studies show a consistent trend of fishing down food webs such that large vertebrates were often targeted and extirpated relatively rapidly. Today, fish are the most commonly identified predator, but sea otters and sea ducks are also important predators in some northern regions (Table 15.1). In all cases, these vertebrate predators are smaller in body size and/or fewer in number than they were 5000–10 000 years ago at first human contact (Steneck *et al.* 2002, 2004). Crustaceans such as spiny lobsters and crabs are among the most important invertebrate predators (Table 15.1). Extrapolating from the known trend in the Gulf of Maine, where extirpation of large predatory finfish led to the dominance of crabs as apex predators, it is possible that crab predation elsewhere is the result of a disrupted trophic cascade. Crabs are dominant predators of sea urchins in Japan and in the UK (Table 15.1), and both regions have a long history of fishing that targeted coastal groundfish. In Europe, coastal groundfish stocks were fought over in the thirteenth and fourteenth centuries, and their depletion is thought to have contributed to the development of distant-water fisheries in Iceland and eventually North America (Kurlansky 1997). It is possible

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that the very early extirpation of apex predators in Europe's coastal zones led to the rise of crabs as predators just as has more recently happened in the Gulf of Maine (Leland 2002).

Overfishing in kelp forests leads to ecological (Estes *et al.* 1989) and possibly absolute (Tegner *et al.* 1996*b*) extinctions. This loss in biodiversity may make these systems more susceptible to invasion from non-native species (Stachowicz *et al.* 1999). Recently, kelp forests in the western North Atlantic have been invaded by the green alga *Codium fragile*, an introduced competitor that could replace the kelp species in this region that have a long history of resilience and dominance (Chapman & Johnson 1990; Levin *et al.* 2002b) (Fig. 15.3b). The possible replacement of *Laminaria* by *Codium* requires sufficient breaks in the canopy for the latter to take hold. This has been facilitated by the introduction in the 1970s of a non-native encrusting bryozoan that coats, embrittles and opens the kelp canopy every summer (Lambert *et al.* 1992; Levin *et al.* 2002*b*). However, these two invaders join a long list of invading species that have become important players in kelp forest ecosystems of the western North Atlantic (Steneck & Carlton 2001). Species such as the common periwinkle (*Littorina littorea*), and the green and shore crabs (*Carcinus maenas, Hemigrapsus sanguineus*) have not only invaded but in many cases have come to dominate the ecosystem (Steneck & Carlton 2001). Whereas other marine systems have a history of invasion, few have seen the large-scale changes in dominance evident in the western North Atlantic. The successful series of invasions there stands in stark contrast with patterns observed in the species-rich southern California kelp forests, where introduced species remain subordinant to native dominants.

Declining water quality

Development on land often reduces the permeability of soil in the watershed, resulting in greater runoff and increased turbidity from plankton and particulates (e.g. Chapters 2, 13 and 16). Where this has occurred, kelp beds have declined because of the shallowing of the photic zone (Edyvane 2003; Valentine & Johnson 2003). Although nitrogen compounds could increase owing to increased supply from land runoff, sewage disposal and atmospheric sources globally, anthropogenic inputs of nutrients are evident only in areas of restricted water exchange. Thus most coastal waters 'appear still to be dominated by inputs from the open ocean' (Jickells 1998).

In heavily urbanized areas of Japan, terrestrial deforestation and damming of rivers is thought to have starved coastal waters of iron and humic substances necessary for kelp development (Suzuki *et al.* 1995; Matsunaga *et al.* 1999). This is hypothesized to have created a phase shift from kelp dominance to coralline dominance without any changes in herbivory, sea temperature or macronutrients (Matsunaga *et al.* 1999). There seem to be no other examples of chemically induced phase shifts from kelps to coralline algae.

Point-source pollution is often very conspicuous but rarely has it resulted in widespread and long-lasting deforestation of kelp ecosystems. In southern California, raw sewage in the 1950s resulted in the loss of kelps and the creation of sea-urchin 'barrens', but these and other impacts were short lived (North 1994) (Fig. 15.3a). The massive 1989 *Exxon Valdez* oil spill in the vicinity of kelp forests of south-west Alaska, however, minimally impacted kelps and, for most components of the ecosystem, full recovery took 2 years or less (Dean & Jewett 2001). Oil coated sea otters in the spill area and, while the impacts are debated, the greatest impact was a decrease from around five otters per km of shoreline to two or three otters per km (Paine *et al.* 1996), and otters may have suffered a long-term and lingering impact from oil in at least parts of the spill area in Prince William Sound (Monson *et al.* 2000). Although thousands of sea otters and seabirds died following the *Exxon Valdez* oil spill, fewer than 10 dead fish were reported (Paine *et al.* 1996). Thus, fish may be inherently less susceptible to oil spills, but are by no means immune.

Given the likelihood that both non-point-source and point-source pollution will rise with increased human population growth, water quality is expected to continue to decline. It is not yet known whether thresholds of accelerated mortality exist with declining water quality as they apparently do for increasing fishing pressure (Myers *et al.* 1997). However, it is currently top–down impacts on ecosystem drivers (such as sea urchins and their predators) that most consistently denude kelp forests.

CONCLUSIONS

Kelps are the largest bottom-dwelling organisms to occupy the euphotic zone. Their size and photosynthesis-to-biomass ratio constrain their distribution globally and locally. Kelp forests fail to develop at high latitudes as a result of light limitations and at low latitudes owing to

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nutrient limitations, high sea temperatures and competition from other macrophytes. On shallow mid-latitude rocky marine shores worldwide, phyletically diverse, structurally complex and highly productive kelp forests develop. These forests are uniquely capable of altering local oceanography and ecology by dampening wave surge, shading the sea floor with their canopies, providing a physical habitat for organisms above the benthic boundary layer and distributing trophic resources to surrounding habitats. In this context, the three kelp forest case studies from North America represent ecosystems along a continuum of natural biodiversity and human interactions. The archaeological literature may be used to estimate an ecological baseline of the structure and function of kelp forests prior to contact with modern humans (Steneck *et al.* 2002) (Fig. 15.2, Table 15.1).

Kelp forests of the eastern Pacific may have facilitated an early coastal migration of humans into the Americas (Erlandson 2002). The concentration and high productivity of vertebrates and invertebrates along this coast would have provided early human settlers with a stable source of food between 15 000 and 10 000 years ago. Archaeological data indicate that coastal settlements exploited organisms associated with kelp forests for thousands of years, occasionally resulting in the localized loss of apex predators, outbreaks of sea-urchin populations and deforestation (Erlandson *et al.* 2004). However, these human impacts on kelp-bed systems were probably localized and relatively ephemeral.

Consumer animals structure kelp forest interactions via two primary drivers, namely herbivory by sea urchins and carnivory from predators of sea urchins, but other forcing functions can be important. For example, kelps are prone to destruction and thinning by storms and competitors. Further, their growth and survival are sensitive to temperature, light and nutrient availability. However, the spatial scale and magnitude of these impacts on kelp forests are small relative to those of consumers. Kelp deforestation worldwide results from sea-urchin grazing, which is controlled by predation in kelp forests where human harvesting impacts have been minimal.

Over the past two centuries, the commercial export of kelp forest consumers led to the extirpation of sea-urchin predators such as the sea otter in the North Pacific and groundfish such as cod in the North Atlantic. In those systems, sea-urchin abundances increased and kelp forests were denuded over vast stretches of coast. In southern California, the high diversities of predators, herbivores and kelp appear to have buffered this system from systemic deforestation.

Biodiversity of kelp forests may also help the ecosystem resist invasion from non-native species. In the species-depauperate western North Atlantic, introduced algal competitors carpet the benthos and threaten the dominance of kelp. Other introduced herbivores and predators have taken hold and have come to dominate components of the system.

Global and regional climate changes have measurable impacts on kelp forest ecosystems. Increasing El Niño frequencies, oceanographic regime shifts and violent storms cause deforestation. This, in combination with the serial loss in biodiversity from overfishing, appears to be the greatest threat to structure and functioning of these systems over the 2025 time horizon.

Kelp forests cannot be considered in isolation. They have vital ecological linkages with deep-sea, subtidal, sandy and rocky intertidal and coastal ecosystems (e.g. Chapters 14, 17 and 22). The high interconnectedness and dependence on kelp organic exports highlight the significant trophic functions these plants play for adjacent ecosystems. This regional trophic connectivity magnifies the importance of kelps to communities well beyond their distribution. While kelp forest ecosystems should persist, they may do so with entirely altered food webs; for example, they may be devoid of key fish and invertebrate species. Some of these alterations will reduce stability not only of kelp forests, but also of neighbouring coastal and marine ecosystems.

Management of kelp forest ecosystems should focus on restoring biodiversity and especially on minimizing fishing on predators. In particular, northern hemisphere species such as sea otters, and the fish sheephead and cod should be restored to fulfil their functional role in the Alaska, California and western North Atlantic systems, respectively. While sea otters are already protected, other commercially valuable species such as Atlantic cod are unlikely to be preserved for the ecosystem role they perform. Southern hemisphere invertebrate predators such as lobsters and wrasses should also be restored to levels where they can function in their ecological roles. Ultimately, human values and political resolve will determine the conservation agenda. Significant investment in education of stakeholders, the general public and policy-makers will be necessary for this conservation goal to succeed.

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